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B.E - (Mechanical) (Sem-VII)(Revised Course 2019-2020)

EXAMINATION JULY 2023

OPEN ELECTIVE Additive Manufacturing

[Time: 3 Hours]

[Max. Marks:100]

Instructions: 1) Answer the questions from Part A, B, and C as per the instruction.

2) Assume any missing data.

3) illustrate your answers with neat sketches wherever required.

Part –A

Answer any **TWO** questions from the following:

- | | | |
|----|---|---|
| Q1 | A) Classify the Additive manufacturing process. | 6 |
| | B) Explain the role of Additive manufacturing in new product development with the aid of flow chart. | 6 |
| | C) Suggest the additive manufacturing process for the Jewellery Manufacturing unit to make patterns for investment casting made up from Liquid Polymer. Justify your answer with the working principle, advantages & disadvantages. | 8 |
| Q2 | A) Explain the different types of materials which can be used to build the prototypes using the additive manufacturing process. | 4 |
| | B) Suggest the Cost-effective additive manufacturing process for the Artificial Jewellery manufacturing Company to develop products from Poly-Lactic Acid (PLA) Justify your answer with the working principle, advantages & disadvantages. | 8 |
| | C) A Medical firm would like to manufacture orthopaedic Implants from Metal powder with an accuracy of 1 micron. Suggest the process with the aid of a neat sketch, working principle, advantages & disadvantages. | 8 |
| Q3 | A) Explain the working of the Thermal Jet Printer with a neat sketch | 4 |
| | B) A Ship repairing company is looking for an Additive manufacturing process which can print five Dimensional objects from Titanium material. Suggest the process with the aid of a neat sketch, working principle, advantages & disadvantages. | 8 |

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- C) An Architectural firm is looking for an Additive manufacturing process which can build prototypes from the laminates. Suggest the process with the aid of a neat sketch, working principle, advantages & disadvantages. 8

Part –B

Answer any **TWO** questions from the following:

- Q4 A) Define Additive manufacturing and explain the need for an additive manufacturing process. 4
- B) Explain with a neat sketch the process of Silicon Rubber tooling 8
- C) Write a note on the factor which influences the accuracy of the additive manufactured component. 4
- D) Differentiate between Direct & Indirect tooling. 4
- Q5 A) Write a note on magic and mimic software used for the Rapid Prototyping. 4
- B) Explain with a neat sketch the process of Aluminium filled epoxy tooling. 8
- C) Write a note on Part orientation and Support generation with a neat sketch. 4
- D) Write a note on object Quadra systems. 4
- Q6 A) Explain the Process of Reverse engineering. 4
- B) Explain with a neat sketch the quick cast process. 8
- C) Write a note on the application of additive manufacturing in the development of Customized implants and prostheses. 8

Part –C

Answer any **ONE** question from the following

- Q7 A) Explain the different process parameters which govern the Mechanical strength of Prototypes made from Laser Engineered Net Shaping (LENS) **6**
- B) Explain with a neat sketch the Process of Sand Casting tooling. **7**
- C) Explain the application of Additive Manufacturing in Computer-aided tissue engineering. **7**
- Q8 A) Explain with a neat sketch the working of Solid Ground Curing Stating its advantages & Disadvantages over other processes. **7**
- B) Explain with a neat sketch the working of Electron Beam Melting Stating its advantages & Disadvantages over other processes. **7**
- C) Explain with a neat sketch the Process of spray metal tooling. **6**